Association of Emitter and Emission Using Deep Learning

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***Abstract*—A method for using deep neural networks (DNN) to perform RF emitter localization is presented. The localization is performed using a DNN to perform dual modality sensor fusion to associate observed RF emissions and detected potential RF emitters. A synthetic dataset is presented as a means of evaluating the validity of the approach in a simplified setting.**

I. Introduction

The problem of locating a radio frequency (RF) transmitter is encountered in various applications. Whether emergency services or commercial applications, high accuracy location estimation is desired.

Many researchers have proposed approaches to exploit different components of RF signals. A survey on recent localization techniques provides a comparison of common range-based array processing approaches, namely: received signal strength (RSS), time of arrival (TOA), time difference of arrival (TDOA), frequency difference of arrival (FDOA), and angle of arrival (AOA) [1]. The benefits and drawbacks of the various range-based approaches are generally understood and variations of the approaches are regularly the focus of research.

TDOA based localization methods have high accuracy but often assume line-of-sight (LOS) between the transmitter and receivers. If the signal propagation environment is non-line-of-sight (NLOS), the accuracy of TDOA based methods is diminished. Some work has been done to improve the accuracy in the NLOS scenarios for TDOA techniques, but mitigating the effot of NLOS environments for TOA techniques has received more attention. Thomas et. al used a biased Kalman filter on the TOA data prior to computing the TDOAs to increase the localization accuracy[2]. An alternative TDOA approach using pilot signals in NLOS environments is presented in [3]. Traditionally, the range-based approaches are unimodal with the RF measurements being the single modality. Deviating from this tradition, the authors of [4] present a multimodal sensor fusion based localization approach that implements an algorithm which uses both video and RF measurements to track a wireless endoscope capsule in an intestine.

Recently, there has been a rise in popularity of deep learning. The creation of datasets such as the ImageNet database and the associated classification challenges [5] has helped fuel the interest, not only by making large datasets available, but because many deep neural networks (DNN) used in the challenges are outperforming the previous state-of-the-art human developed algorithms. Unrelated to source localization, researchers are using DNNs to jointly process audio and video data in an effort to teach a computer to read lips [6].

A multimodal localization technique is presented in this paper as an alternative to the unimodal approach. The proposed method accomplishes source localization by associating RF emissions with identified potential emitters. The association is determined using DNNs to perform the fusion processing of an array of antennas and standard imagery. This work is introductory and is intended to provide an initial understanding of the viability in using DNNs as an alternative to traditional range-based localization approaches.

The development of the multimodal fusion approach using DNNs is the focus of the remainder of this paper. An overview of the problem is presented in section II. A brief discussion of TDOA localization is given in section III. The proposed DNN, dataset, and association process is discussed in section IV. Results are presented in section V-C prior to the conclusion in section VI.

II. Problem Overview

A. Experiment Setup

An overhead view of the experimental setup is shown in figure 1. The field of view of one or more cameras, each having an independent pointing direction, establishes a fixed region of operation. Three antennas with known, fixed locations (x_1, y_1, z_1), (x_2, y_2, z_2), and (x_3, y_3, z_3) are located around the region of operation. Multiple potential RF emitters exist both in and around the region of operation. In figure 1, the potential emitters are depicted as vehicles for simplicity. The true number of emitters is not assumed to be known. Each antenna observes each RF emission and at this stage of research, the RF propagation is assumed to be LOS with no multipath component present.

B. Association Problem

Within the scope of this paper, the basic association problem is that of determining which potential emitter in the region of operation is the true source of an observed RF emission. In the presence of multiple RF emissions the problem requires

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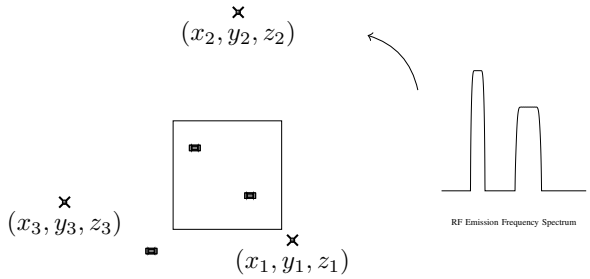

Fig. 1: This is the simulation environment. The region of operation and the cars lie in the same plane.

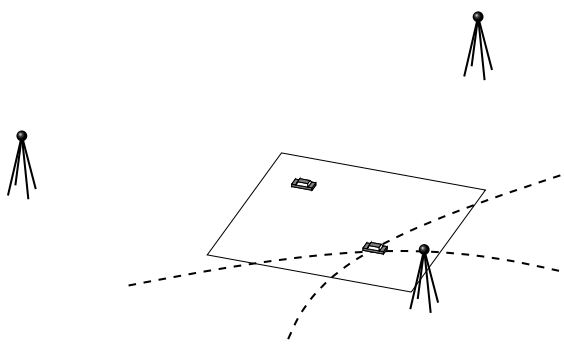

Fig. 2: The intersection of two isochrones for an emitter in the region of operation. The isochrones are the dashed lines and lie in the same plane as the region of operation.

the determination of multiple associations. Additionally the true emitters are not required to reside within the region of operation, thus allowing for observed RF emissions for which there is no true association with any identified potential emitters.

III. Simplified TDOA Localization

In an isotropic propagation enviroment, calculation of a TDOA between two receiving antennas yields two paths of constant time difference called isochrones. The two isochrones are symmetric about the midpoint between the two antennas. Because the isochrones represent constant TDOAs, the emitter may be positioned anywhere along either isochrone and the position is ambiguous. Including additional antennas provides additional independent TDOAs and corresponding isochrones. The ambiguity from a single TDOA is removed because the emitter must be located at the intersection of the isochrones. The intersection of two isochrones for an emitter in the region of operation is shown in figure 2. The isochrones which intersect outside of the region of operation are not shown.

IV. DNN Approach

A. Solution Architecture

The two modalities used in this work are RF measurements and RGB images. Preprocessing on each modality is assumed to be available and implemented to reduce the number of layers and the number of neurons in each layer of the DNN. It is assumed that the optical preprocessing is able to identify (x, y) coordinates of potential emitters in the region of operation. Additionally, the RF prepocessing is assumed to produce two independent scalar TDOA values from the RF measurements made at the three antennas. The outputs of the preprocessing stage are concatenated to create a four element input vector for the DNN. The DNN, the focus of this paper, is trained to compute a probability based on the four element input vector. The system overview is shown in figure 3. The target probability is defined in section IV-B.

B. Dataset Information

Every input sample into the DNN, as mentioned in section IV-A, is a four element vector $[x, y, TDOA_{01}, TDOA_{02}]^T$. The input vector is a pairing of a single possible emitter location and single set of computed TDOAs. Let a valid pairing be defined as an input vector where the position (x, y) is the location of the emitter which yielded the TDOAs. Similarly, let an invalid pairing be defined as any input vector where the position (x, y) is not the location of the emitter which yielded the TDOAs.

The DNN has no knowledge of the emitter location and must either classify the input vector as a valid or invalid pairing. Rather than train a DNN to perform the binary classification, the DNN was trained to provide a confidence level in the pairings validity.

In this research the DNN was trained to provide a confidence level by outputting the probability that the input position (x, y), referred to as $\mathbf{x}_{test}$, is the true emitter's position, $\mathbf{x}_{true}$. The training was accomplished using a dataset containing valid pairings of positions and TDOAs where the target probability is 1, and invalid pairings where the target probability is given by

$$P(\mathbf{x}_{test} = \mathbf{x}_{true} | TDOAs) = e^{-\alpha ||\mathbf{x}_{test} - \mathbf{x}_{true}||_2} \quad (1)$$

where

$$\alpha = -\frac{1}{d} \times ln(p) \quad (2)$$

is the desired probability, p, at a separation distance, d, between the true RF emitter position, $\mathbf{x}_{true}$, and the position of interest, $\mathbf{x}_{test}$. The larger the value of α, the closer $\mathbf{x}_{test}$ and $\mathbf{x}_{true}$ must be before the DNN will output a significant confidence level. Equation 1 for varying levels of α is plotted in figure 4.

The simulated dataset used for this paper is generated using algorithm 1 with $\alpha = 0.138$ and $N = 25$.

C. DNN Architecture and Training

The DNN used is a fully connected neural network and the architecture is given in table I. The seven hidden layers all use the rectified linear unit (relu) as the activation function and the single output neuron uses the sigmoid function as it's activation function.

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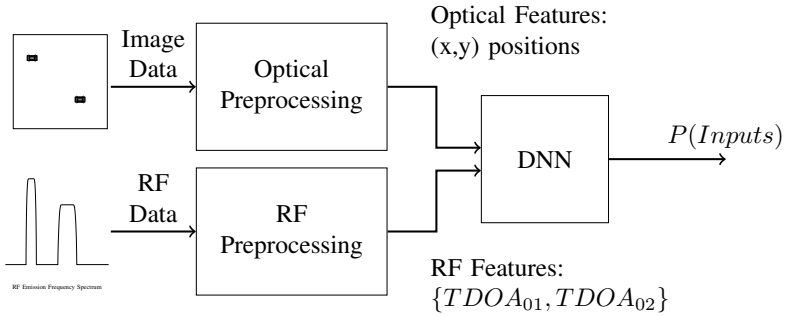

Fig. 3: Overview of the system processing architecture. Image preprocessing produces (x,y) positions of potential emitters. RF preprocessing computes TDOAs for the observed RF emissions. The (x,y) positions and the two independent TDOAs are concatenated and input into the DNN. The output of the DNN is a probability value described in section IV-B. The DNN block is the focus of this paper.

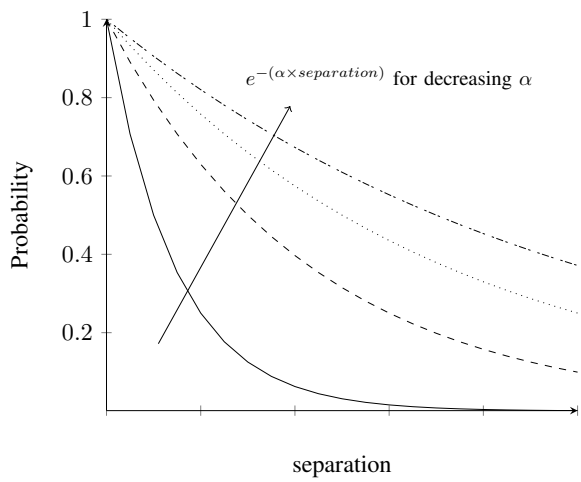

Fig. 4: The desired DNN output behavior as the physical separation between a location of interest and the true emitter location. A larger α prevents the DNN from being as confident in a pairing's validity as a smaller α.

TABLE I: Architecture specification for the trained DNN models used in this paper.

Layer	Neurons	Activation
Input	4	N/A
Hidden 1	50	relu
Hidden 2	60	relu
Hidden 3	50	relu
Hidden 4	50	relu
Hidden 5	50	relu
Hidden 6	60	relu
Hidden 7	50	relu
Output	1	sigmoid

Algorithm 1 Algorithm for generating the synthetic dataset used in this paper.

```
Initialize sensor array geometry
Identify region of interest
Specify α
Set N number of incorrect samples for every correct sample
for k = 1 : Number of input samples do
    Randomly select position for every correct sample
    Compute the TDOAs for every correct sample
    Set target probability for every correct sample to 1
    for Every correct sample do
        Randomly select N positions for incorrect samples
        Pair incorrect sample with correct sample's TDOA
        Compute probability for position and TDOA pair
    end for
end for
Use appropriate scaling and dataset normalization
Write to HDF5 file
```

The model was trained using tflearn [7]. The mean squared error of the DNN output probability and the target probability was minimized using Adam [8].

D. Making Decisions

The output of the DNN is a probability and as such, has a value in the range $[0, 1]$. Traditional hypothesis testing is used to make a decision on the validity of the input vector pairing. If the set V is defined to be the set of valid input vectors $\mathbf{x}$ (valid input vectors are still defined to be the true position, (x,y), and TDOA pairings) the two competing hypothesis are

$$\begin{aligned} H_0 &: \mathbf{x} \notin V \\ H_1 &: \mathbf{x} \in V \end{aligned} \tag{3}$$

The null hypothesis is that the observed TDOAs did not result from an RF emission at the position (x, y) in the input vector $\mathbf{x}$. The alternative hypothesis is that the observed TDOAs did result from an RF emission at the position (x, y) in the input vector $\mathbf{x}$. Setting up a likelihood ratio test and simplifying

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(recognizing $P(\mathbf{x} \in V) + P(\mathbf{x} \notin V) = 1$), the following decision rule is obtained:

$$\phi(\mathbf{x}) = \begin{cases} \mathbf{x} \notin V & P(\mathbf{x}) \geq \tau' \\ \mathbf{x} \in V & P(\mathbf{x}) < \tau' \end{cases} \tag{4}$$

where the value τ' is a threshold value.

Decisions made using the rule given in equation 4 are evaluated using traditional definitions of the probability of false alarm and the probability of detection:

$$\begin{aligned} P_{FA} &= P(\text{decide } \mathbf{x} \in V \mid \mathbf{x} \notin V \text{ is true}) \\ P_D &= P(\text{decide } \mathbf{x} \in V \mid \mathbf{x} \in V \text{ is true}) \end{aligned} \tag{5}$$

V. Results

Multiple approaches were implemented to evaluate the DNN's performance. The receiver operating characteristic (ROC) curve is plotted to evaluate the DNNs ability to provide the probability used in identifing valid input vectors. The performance of the model is dependent upon the location of the true emitter in the region of operation. This behavior is observed and evaluated by visualizing the expected error and learned response throughout the region of interest.

A. ROC Curves

The ROC curve is constructed by evaluating the decision rule in equation 4 and sweeping the value of τ' through the range $[0, 1]$ and then estimating the probabilities in equation 5. 50,000 test points comprised of 25,000 valid pairs and 25,000 invalid pairs are evaluated with the model for every value of τ'. The 25,000 invalid pairs are generated by randomly sampling a position and then calculating the TDOAs, perturbing the true position by a small amount in a random direction, and pairing the new positon with the TDOAs. The 25,000 valid pairings are generated in a similar manner, with the exception that no position perturbation occurs. The entire set of valid and invalid pairs is then shuffled.

Decisions about a pairings validity are made according the decision rule given in equation 4 and then compared to the true status provided during dataset generation. The number of valid pairing input vectors correctly identified as valid is the number of correct detections. The number of invalid pairing input vectors incorrectly identified as valid is the number of false alarms (keeping with traditional hypothesis testing). The probabilities are estimated by dividing both the number of correct detections and false alarms by 50,000.

Four separate curves are shown in figure 5. The different curves were obtaining by making decisions of various difficulty using the DNN. The difficulty is changed by adjusting the amount of position perturbation used when generating the invalid pairings. The four levels' perturbation ranges in meters are: level 1- 4-7 (m), level 2- 3-6 (m), level 3- 2-5 (m), level 4- 1-3 (m). For a perfect detector, there exists a value of τ' where $P_D = 1$ and $P_{FA} = 0$. Three of the curves approach this behavior while the remaining curve begins making false alarms in an effort to correctly identify more valid pairs. This behavior is expected because as the amount of perturbation increases, the decisions should become easier.

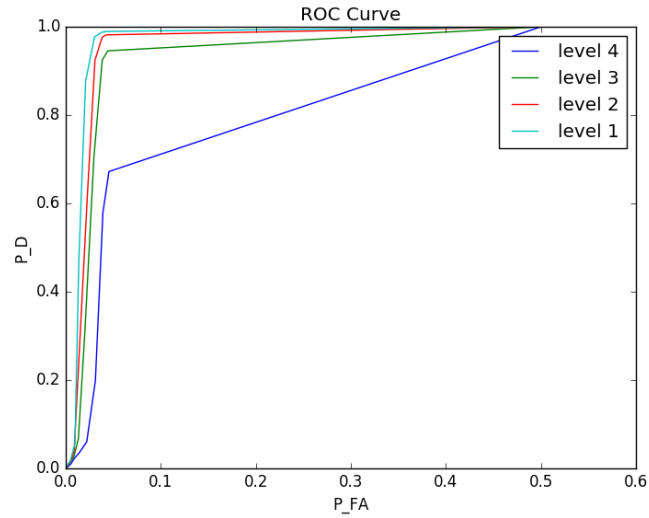

Fig. 5: The ROC curve produced using the rule in equation 4. The varying levels are described in section V-A.

B. Location Dependent Expected Error

Accuracy of traditional TDOA based localization algorithms is dependent on the sensor configuration. The accuracy changes as the RF emitter location changes because it's relation to the sensors also changes. In this work, similar challenges to the accuracy are manifest by evaluating the model throughout the entire region of operation. For an emitter at any arbitrary location within the region of operation and its corresponding TDOAs, the model may return a probability indicating that the source is located elsewhere. Let the distance between the predicted location and the true location be defined as the location dependent expected error (LDEE).

The LDEE is beneficial when it is visualized over the region of operation. A coarse grid of locations is selected as testing points. For every test point, the TDOAs are computed and then paired with points chosen using a fine grid of locations surrounding and including the test point. After the DNN model evaluates every pairing, the distance between the test point and the point with highest probability is calculated and stored as the LDEE at that test point. After the evaluation occurs at every test point, the LDEE over the entire region of operation is visualized by color coding the magnitude of the error. The LDEE for this paper's DNN is displayed in figure 6.

The region of operation in this instance is roughly 350 meters by 280 meters. The magnitude of the error is also given in meters. The worst case error is roughly seven meters and occurs near the corner of the region of operation. Away from the boundaries, the error is expected to be less than two meters. The irregularity of the expected error easily confirms the accuracy of the DNN is location dependent.

C. Learned Response

The target probability given by equation 1 and represented in figure 4 is two dimensional. To evaluate the trained model's

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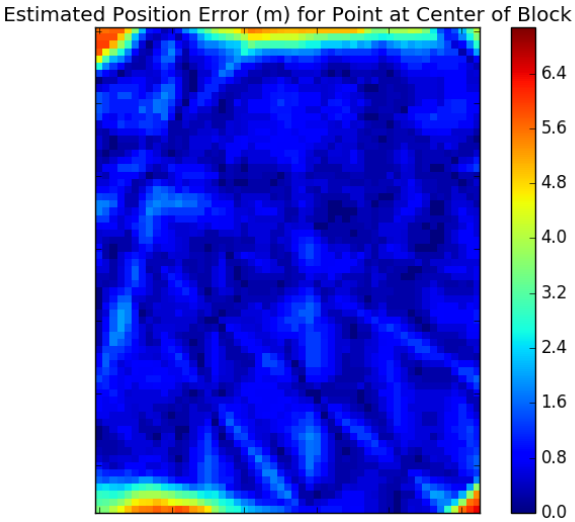

Fig. 6: The location dependent expected error given in meters is represented by colors. The corners in the region of operation are areas where the highest expected error occurs. The region is roughly 350 meters by 280 meters.

ability to learn the target probability output, a series of images is generated. A dataset is prepared for evaluation with the trained DNN model by selecting a location to be an RF emitter, calculating the TDOAs, and pairing the TDOAs with densely sampled points throughout the region of operation. Storing the DNN outputs and creating an image with the probabilities represented using color demonstrates that the DNN is learning the desired response.

In the dataset, only one pairing is valid and it is seen in the region of the image where the probability is nearly $P(Input) = 1$. Every other pairing in the dataset is invalid and the DNN's output probability decays rapidly. A selection of visualizations is presented in figure 7. Visualizations for RF emitters at different locations are included to demonstrate the location dependent accuracy of the model. The response near the boundaries elongates and provides further evidence of the location based behavior. The region of operation is once again roughly 350 meters by 280 meters.

VI. Conclusion

In this paper, an approach using DNNs for localization by associating potential emitters and emissions was presented. A synthetic dataset was generated to evaluate the viability of using DNNs for this task. The work presented in this paper is introductory and the results validate the continued research of DNNs abilities in less ideal scenarios. Future work will remove the isotropic propagation environment assumption and include fading, multipath, and NLOS. Lastly, throughout this work, the antenna geometry remained fixed and future work may also incorporate a variable geometry.

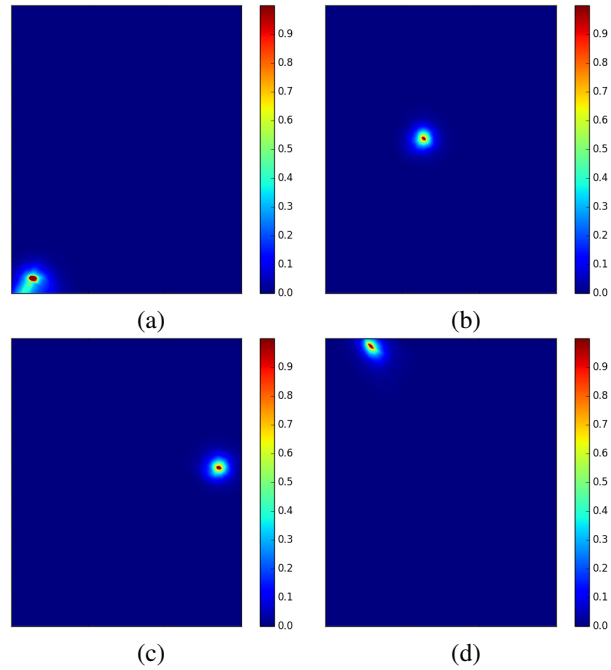

Fig. 7: Visualization of the trained model's learned output throughout the entire region of operation for a few instances of a single RF emitter. The location dependent behavior is apparent in (c) and (d). The color indicates the DNN's output probability.

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